

INTRODUCTION

1. Data Source:

The dataset was obtained from Excelerate Talent Discovery Online Platform. The dataset is an Excel (.xlsx) file, which tracks learners' journey from the sign-up stage through application behavior, demographics, institutions, and opportunity timelines. The data has already undergone cleaning and feature engineering, making it suitable for trend analysis and relationship discovery.

2. Dataset Summary:

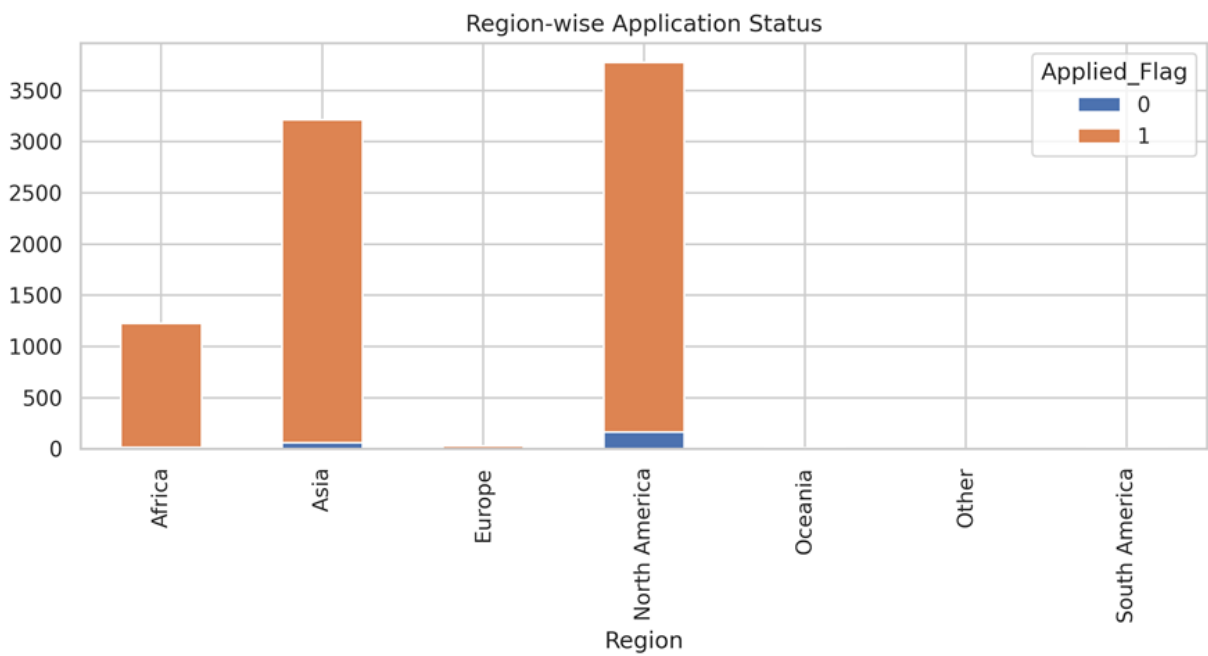
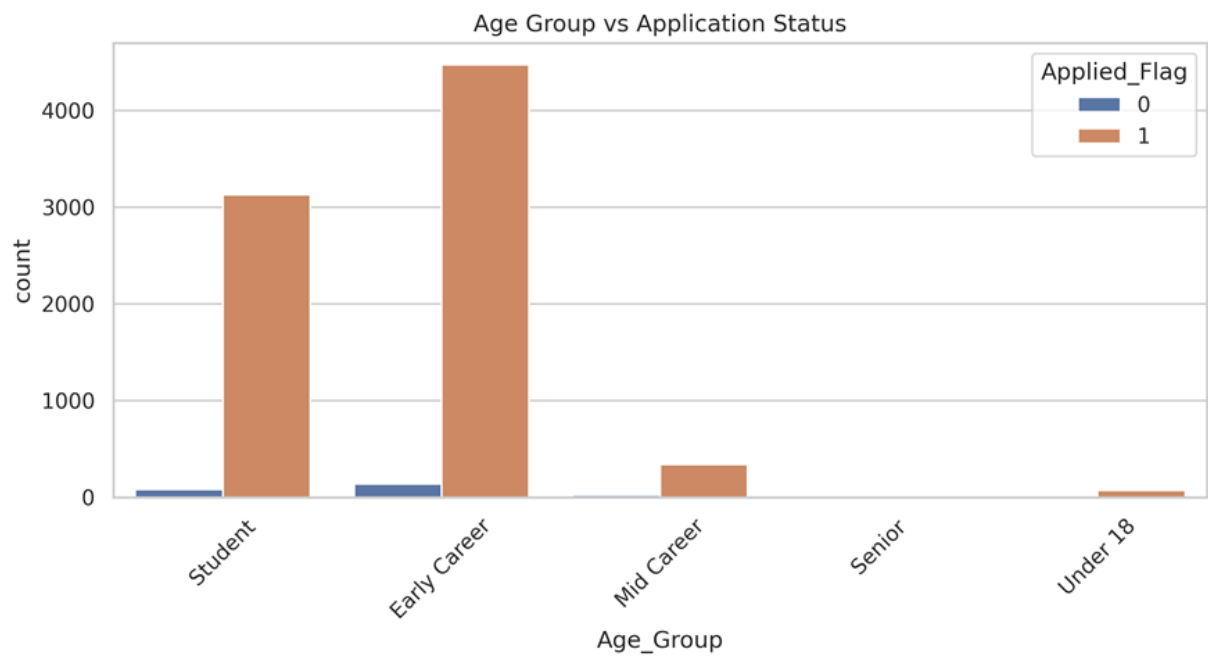
- The dataset originally contains 8263 rows and 27 columns.
- The columns in the dataset are: Learner_SignUp_DateTime, Opportunity_Id, Opportunity_Name, Opportunity_Category, Opportunity_End_Date, First_Name, Date_of_Birth, Gender, Country, Institution_Name, Current/Intended_Major, Entry_created_at, Status_Description, Status_Code, Apply_Date, Opportunity_Start_Date, Learner_Age, Age_Group, Opportunity_Duration_Days, Signup_Month, Signup_Year, Risk_Outcome, Region, Applied_Flag, Signup_Entry_Lag_Days, Ongoing_Opportunity_Flag, Signup_Apply_Delay_Days_Filled
- The dataset has no missing value.

3. Analysis Goals:

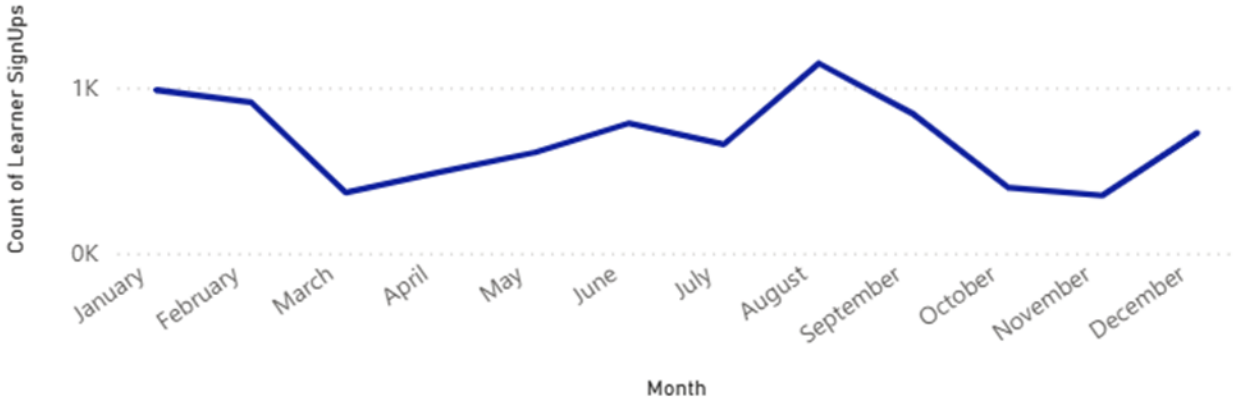
The goals of this analysis are;

1. To know which opportunities attract the most signups
2. To know which opportunities are completed successfully, and
3. To identify patterns that lead to drop-offs.

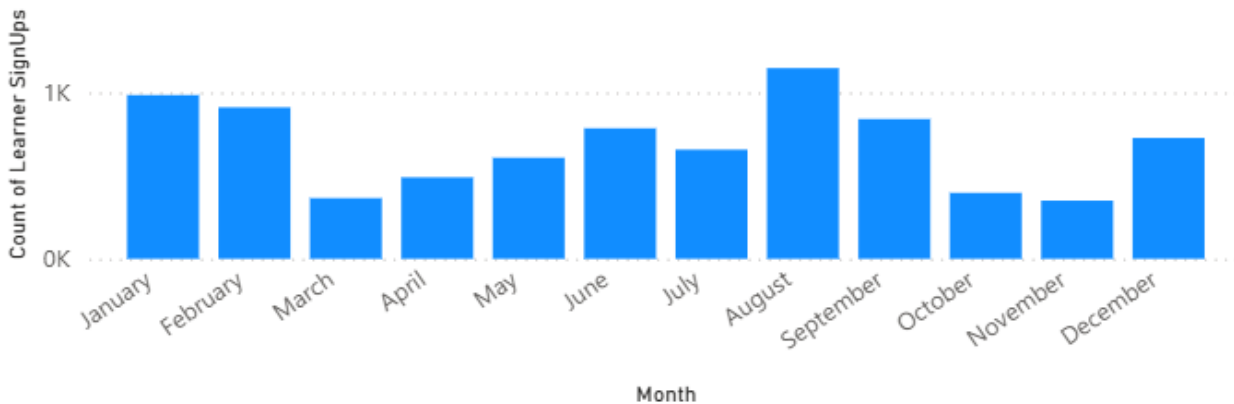
Signup Trends



Count of Learner SignUps by Month



Count of Learner SignUps by Month



1. Signup Rate Highlights

- **Annual Peak:** The highest volume of signups occurred in August, exceeding the 1K mark.
- **Strong Start:** The year began with high volume in January (approx. 1,000) before experiencing a gradual decline.
- **Late Recovery:** After a stagnant autumn, there is a sharp upward trend moving into December, suggesting a strong year-end push.
- **Age Group:** Most signups come Early Career and student categories
- **Region:** Most applicants are from the Northern America and Asia region while a significant number also comes from Africa region.

2. Key Trends & Volatility

- **The Q1 Drop:** There is a sharp decline from February to March, where signups hit a significant local low.
- **Mid-Year Climb:** From March through June, the chart shows a steady, healthy increase in signups.

- **The Q4 Slump:** Following the August peak, there is a consistent three-month decline, with November representing the lowest point in the latter half of the year.

Completion Trends



1. Completion Rate Highlights

The completion rate was defined by two primary cycles of activity:

- **Primary Peak:** The highest level of productivity occurred in **February**, reaching **5 completions**.
- **Secondary Peak:** A strong recovery was noted in **September**, reaching **4 completions**.
- **Baseline Consistency:** For 25% of the year (March through May), the output remained perfectly stable at **2 completions per month**.

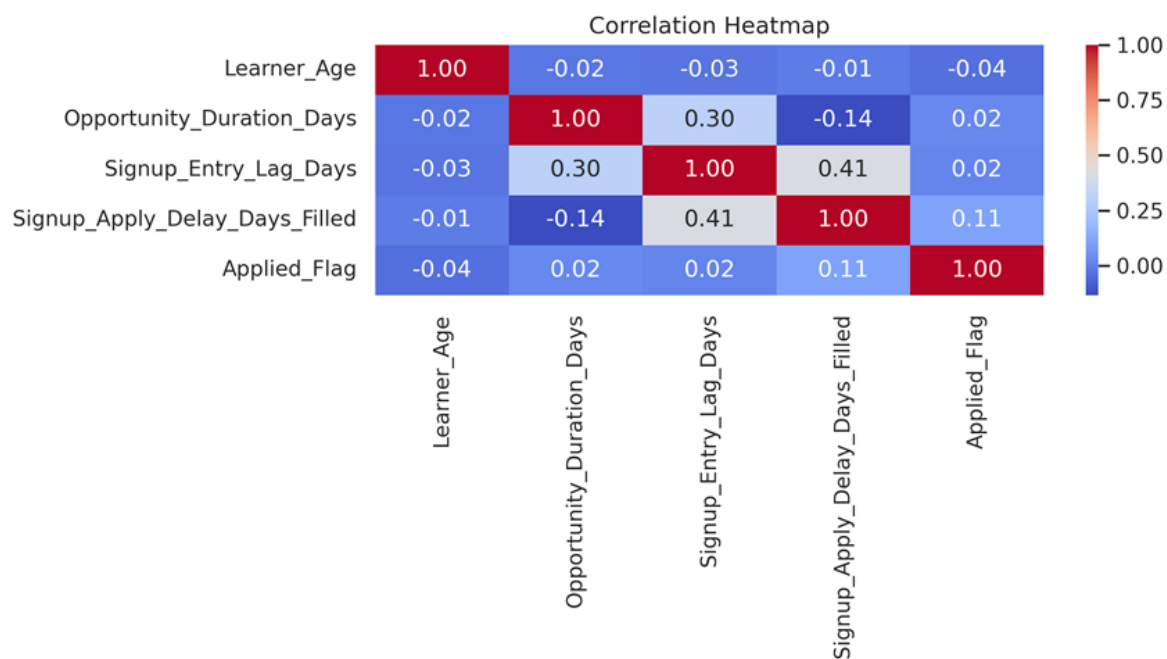
2. Critical Downturns & Volatility

The most significant area of concern is the **July Slump**. After a strong performance in June (4 completions), the count dropped by **75%** to an annual low of **1 completion** in July. This "spike and crash" pattern suggests that seasonal factors are heavily impacting completion rate.

3. Quarterly Trend Overview

- **Q1 (Growth):** Started with moderate rate of completion, peaking rapidly before stabilizing.
- **Q2 (Plateau):** Characterized by steady, predictable output.
- **Q3 (Volatility):** Experienced both the annual low (July) and a significant recovery (September).
- **Q4 (Contraction):** A steady downward trend from September through December, ending the year at the same level it began.

Patterns and Correlations



1. Demographics vs Engagement

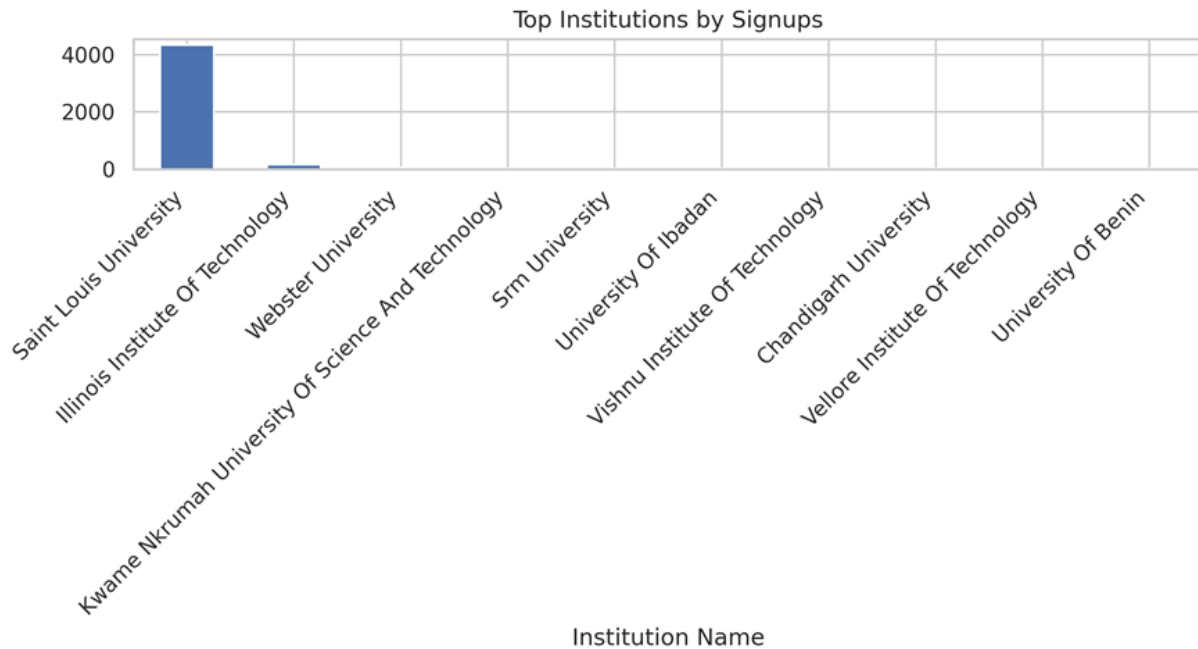
Analysis across **Age_Group**, **Gender**, and **Region** reveals meaningful behavioral patterns.

Insights:

- *Students* form the largest and most engaged group.
- *Early Career* learners show higher application follow-through compared to other groups.
- North America and Asia contribute the highest volume of signups, with relatively strong engagement levels.

1.1 Institution-Level Patterns

Institution names show repeated participation from universities such as *Saint Louis University* and *Illinois Institute of Technology*.



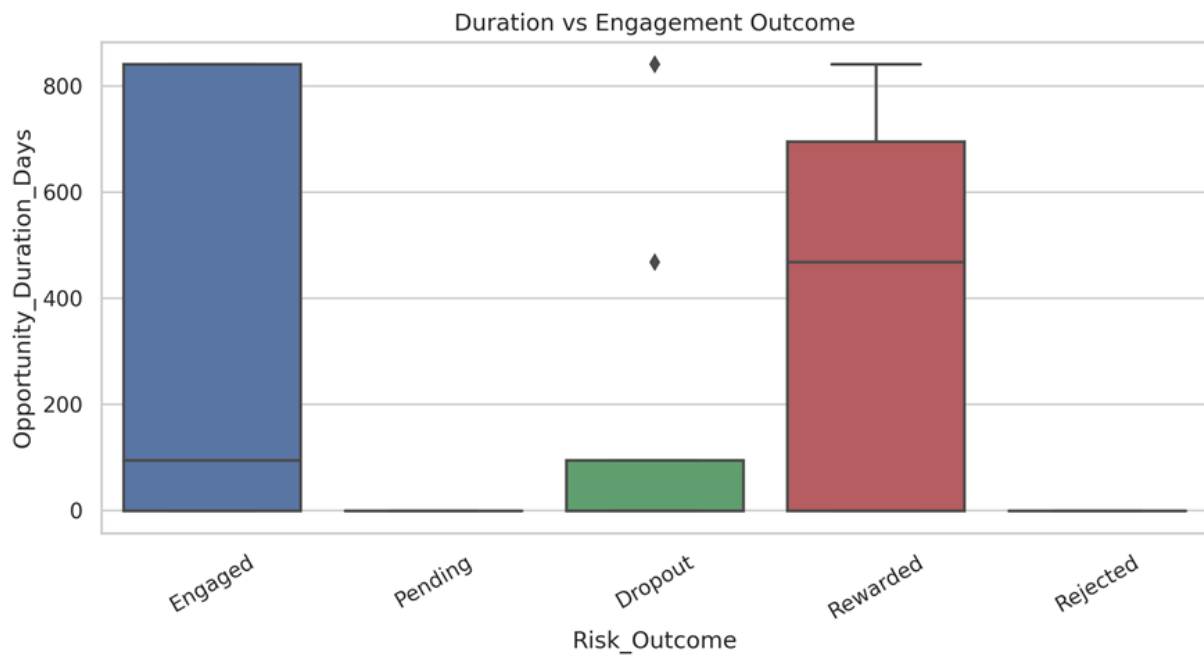
1.2 Insights:

- Certain institutions consistently produce higher engagement rates.
- Institutional partnerships may significantly influence completion outcomes.

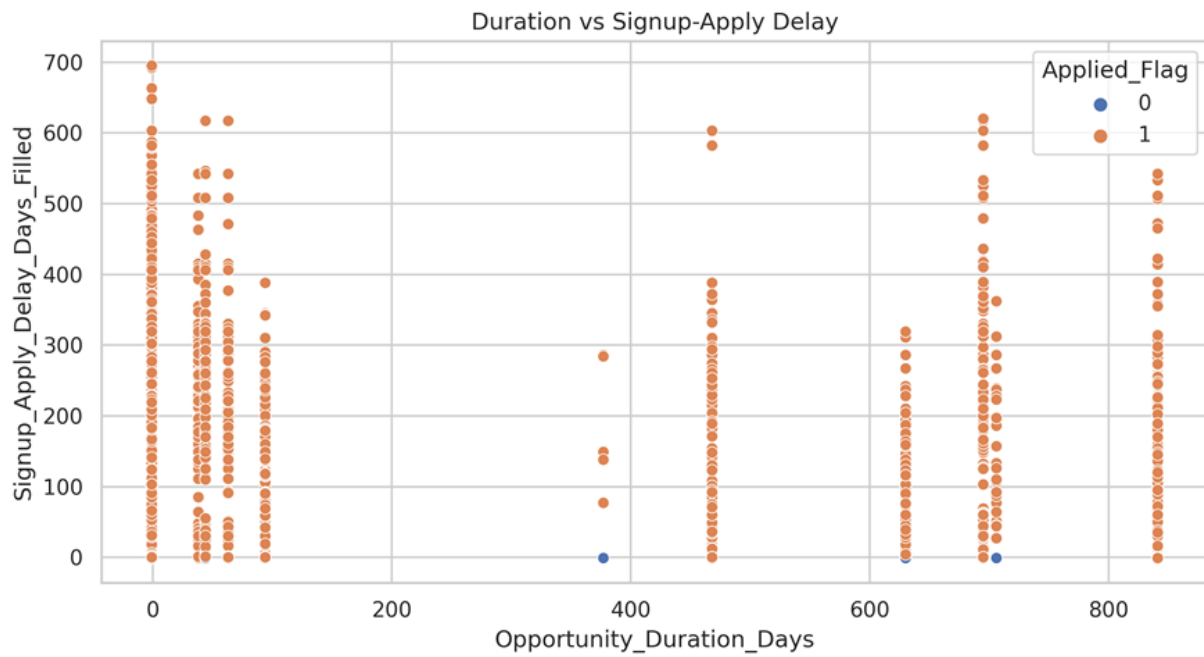
1.3 Opportunity Duration and Completion

The variable **Opportunity_Duration_Days** was analyzed against engagement indicators.

Duration vs Engagement



Duration vs Delay



Insights:

- Extremely long-duration opportunities tend to show reduced application follow-through.
- Moderate-duration opportunities demonstrate the highest engagement, indicating an optimal length for learner commitment.

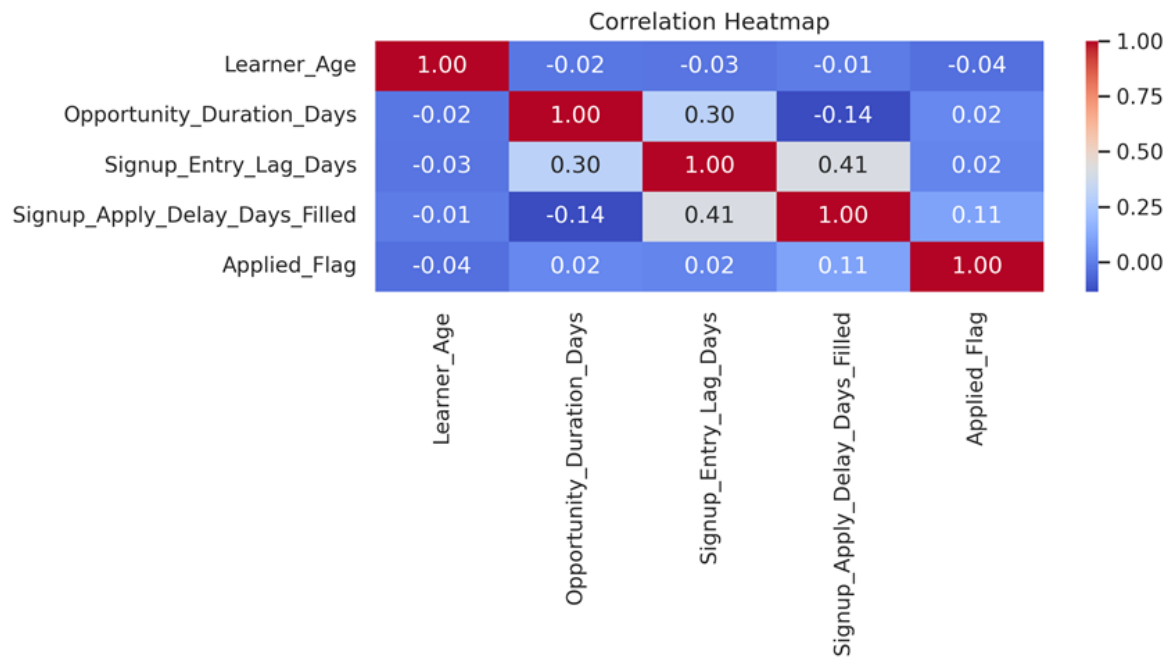
2. Correlation Analysis

A correlation matrix was generated using numeric features such as:

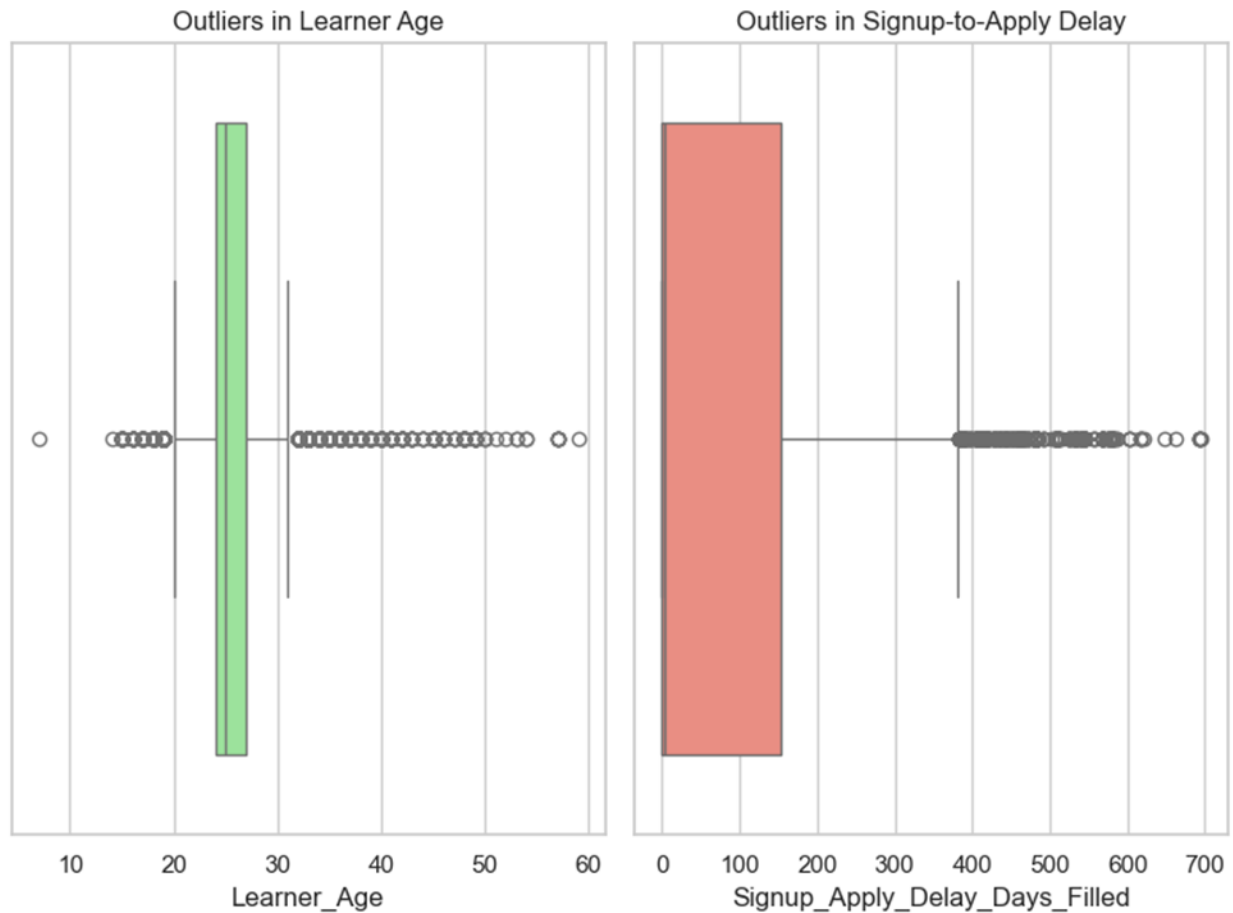
- Learner_Age
- Opportunity_Duration_Days
- Signup_Entry_Lag_Days
- Signup_Apply_Delay_Days_Filled
- Applied_Flag

Key Correlations:

- Negative correlation between application delay and application success.
- Mild correlation between opportunity duration and engagement outcome.



OUTLIERS AND ANOMALIES



1. Understanding Learner Age (The "Lifelong Learner" Segment)

The first chart focuses on the age of users. While the platform is primarily designed for students and early-career professionals, the data shows a significant "tail" of older users.

- **The "Normal" Range:** Most users fall between the ages of **18 and 31**. This is our core demographic.
- **The Outliers:** We identified nearly **900 users** who are over the age of 31, with some reaching as high as 60.
- **What this means for us:** These are not errors. These represent a secondary market of "Non-traditional Students" or "Lifelong Learners." They are statistically different from the main group, suggesting we might need a separate marketing or content strategy specifically for mature professionals looking to upskill.

2. Signup-to-Apply Delay (The "Dormant User" Segment)

The second chart measures how long it takes for a user to apply for an opportunity after they first sign up for the platform.

- **The "Instant" Users:** The vast majority of users are "High-Intent." They sign up and apply for an opportunity on the **very same day**. This shows that initial landing pages are highly effective at converting interest into action.
- **The Outliers:** We found about **300 users** who waited over a year (380+ days) to apply after signing up.
- **What this means for us:** This is a great sign for long-term retention. It shows that users who joined a long time ago are still checking their emails or returning to the platform when a new, relevant opportunity arises.

3. Identifying System Anomalies (The "Bugs")

In this specific analysis, I applied a filter to remove "Negative Delays." This is where we found our true **Anomalies**.

- **The Issue:** Approximately **974 records** showed users applying for an opportunity *before* they actually signed up.
- **What this means for us:** This is a logical impossibility and points to a technical "Anomaly." It is likely a system bug in how timestamps are recorded when a user uses a "Quick Apply" feature or a result of a data migration error.

Recommendations

1. For Marketing & User Acquisition:

- Leverage Peak Seasons: Capitalize on the high-intent periods of January and August with targeted marketing campaigns and opportunity launches to maximize signups.
- Mitigate Seasonal Drops: Develop proactive engagement strategies (e.g., email nurture sequences, highlighted opportunities) for the observed slump periods (February-March and September-November) to maintain momentum.
- Segment and Target: Tailor outreach for the high-performing segments:
 - Students and Early Career individuals in North America and Asia.
 - Develop specific strategies for the identified "Lifelong Learner" segment (users over 31) to better serve their upskilling needs.
- Strengthen Institutional Partnerships: Deepen ties with high-engagement institutions like Saint Louis University and Illinois Institute of Technology, and use their success as a model to onboard similar institutions.

2. For Product & Opportunity Design:

- Optimize Opportunity Duration: Structure opportunities with a moderate duration, as analysis shows extremely long durations reduce application follow-through. Use data to define the "sweet spot" for engagement.
- Improve the "Quick Apply" Journey: Streamline the path from signup to application to convert the high volume of same-day, high-intent users more effectively.

- **Prioritize Bug Fixes:** Immediately investigate and resolve the critical data anomaly where users appear to apply before signing up. This impacts data integrity and could mask usability issues with features like "Quick Apply."

3. For User Engagement & Retention:

- **Reactivate "Dormant Users":** Implement a re-engagement campaign targeting users with a long Signup-to-Apply Delay (e.g., those waiting 30+ days). Personalized emails about new or relevant opportunities can effectively bring them back.
- **Reduce Application Delay:** Since a negative correlation exists between delay and application success, introduce features like reminders, opportunity milestones, or simplified application steps to encourage faster follow-through after signup.
- **Develop Quarterly Engagement Plans:** Align community and support activities with the observed quarterly trends (e.g., focused support in volatile Q3, celebratory content during Q1 growth).

4. For Data Governance & Analysis:

- **Implement Data Quality Checks:** Establish automated checks to flag future anomalies like negative time delays, ensuring ongoing data reliability.
- **Deep-Dive on Drop-Offs:** Conduct a focused analysis on users who sign up but never apply (Applied_Flag = 0) to understand the specific points of friction.
- **Track Completion Drivers:** Move beyond correlation to analyze the causal factors behind successful completions, especially for opportunities that buck the trend during low periods like July.

Conclusions

This analysis successfully identified key trends and patterns within the learner journey dataset. The platform effectively attracts its core demographic of students and early-career professionals, with clear seasonal peaks in signups (January, August). Engagement is strongly influenced by opportunity duration and user demographics, with certain institutions and regions being standout performers.

Two significant user segments beyond the core were identified: valuable "Lifelong Learners" and "Dormant Users" who represent a retention opportunity. The most critical finding is a system anomaly indicating flawed timestamp data for nearly 1,000 records, which must be addressed to ensure accurate analysis.

Ultimately, the path to improving completion rates lies in a dual strategy:

- 1) Optimizing the product and opportunities based on behavioral insights (like ideal duration).
- 2) Executing targeted, data-driven marketing and engagement campaigns focused on high-potential user segments and seasonal cycles. Addressing the data quality issue is the essential first step to building all future decisions on a trustworthy foundation.